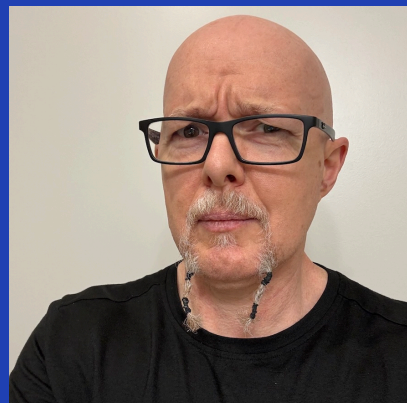


Platform SSO: You're Holding It Wrong!



MARC L. GRUBB

Sr. Mac Platform Engineer
IS Enterprise Engineering
Sophos, LTD

"Helping Enterprise
leverage Macs
to make awesome
people awesomer"



Trigger Warnings

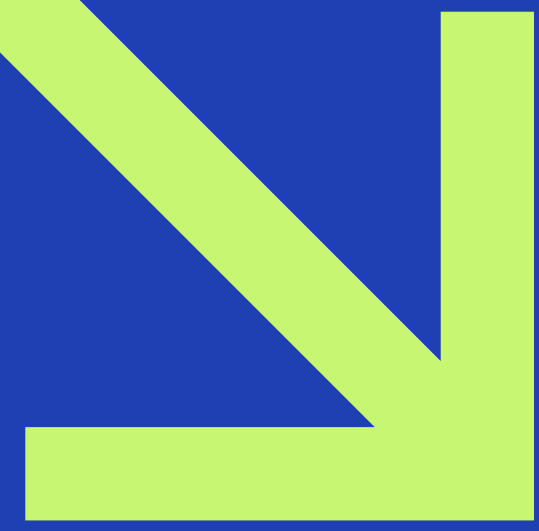
I'm going to say "Macs" as the plural form of "Mac".

The Apple Style Guide wants me to say "Mac computers" or "Mac devices". IMO no one should ever use those phrases. They are impersonal, and we love our Macs.

Also, this presentation may contain mild yodeling.

Passwordless
Password Sync
Phishing-Resistant
FIDO
Hello?

How exactly do our Macs fit within an Entra ID world?



AGENDA

1. Understanding "Passwordless"

2. Learning how Macs exist within a passwordless Microsoft Entra ID environment

3. Understanding Passkeys

4. Learning how to safely test and deploy passkeys in production

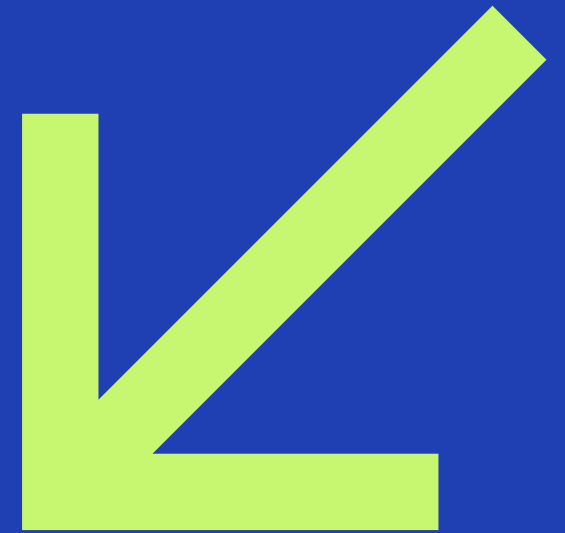
but first...



They say an image is worth
a thousand words...
but a song is...
um, maybe a hundred words?
but it's just as memorable.

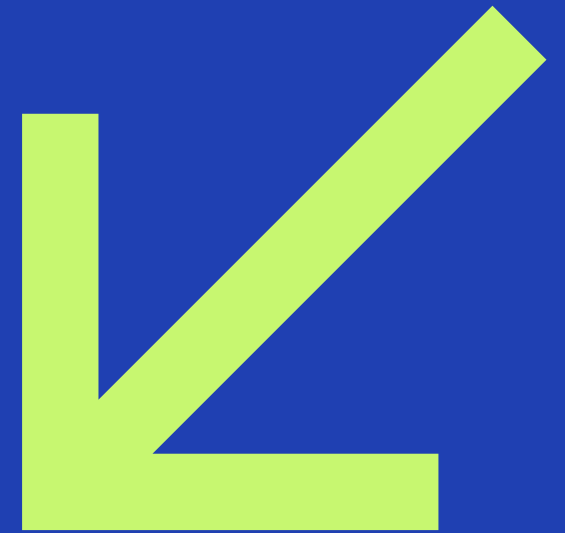
“

*I knew a misnamed man called Lucky
bought a coffee from his friend Starr Bucky*



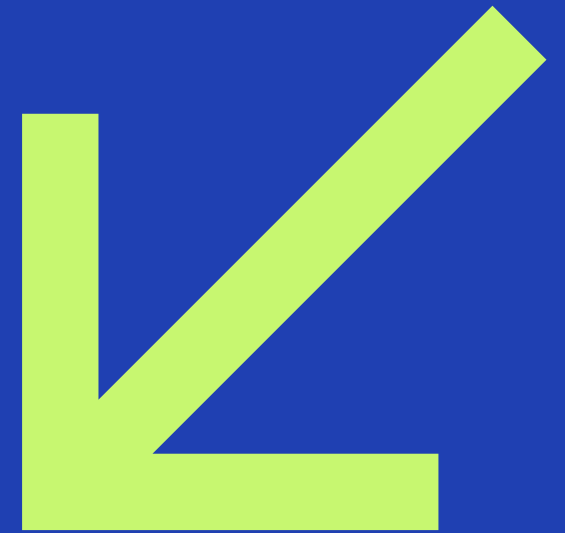
“

the order was ready - so they they called his name



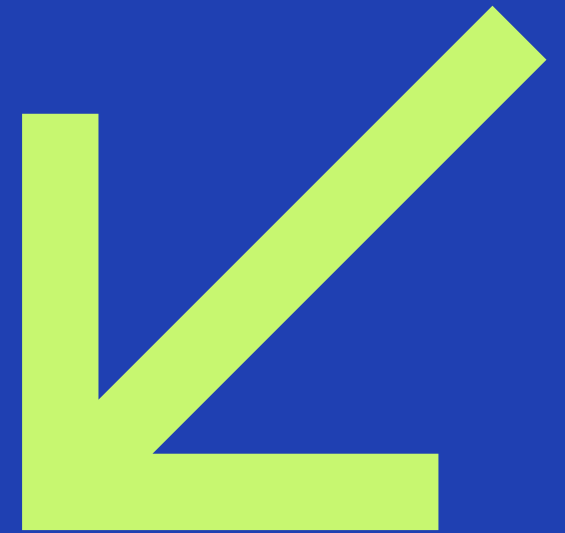
“

*the cup was hot, so he grabbed the lid
there's no Command-Z for the thing he did*



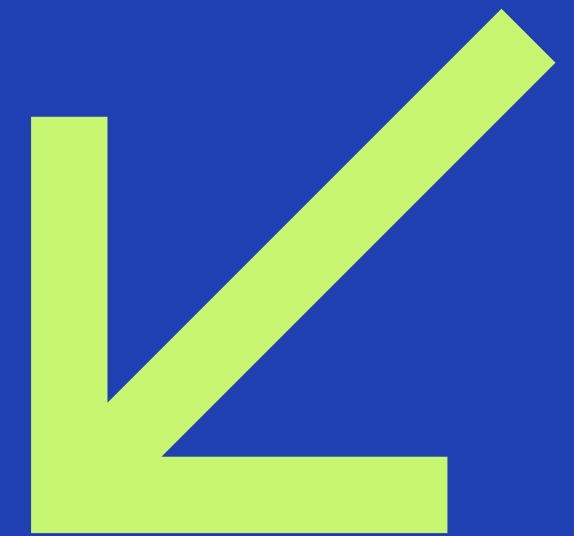
“

poor Lucky figuratively died of shame



“

He was holdin' it wrong
He was holdin' it wrong

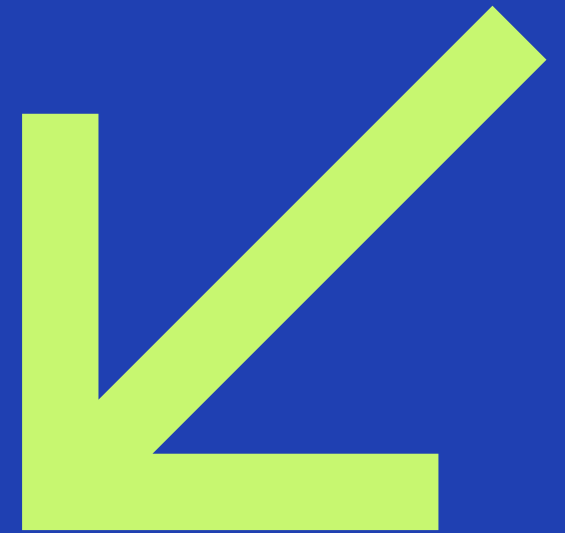


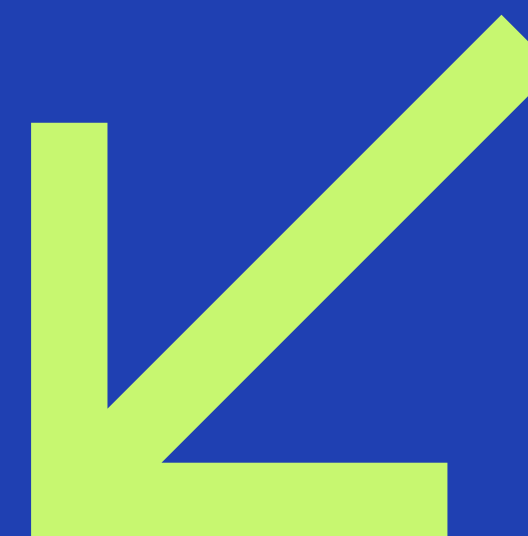
“

I think its fair - to get all scoffy

He shoulda realized he'd drop his hot coffee

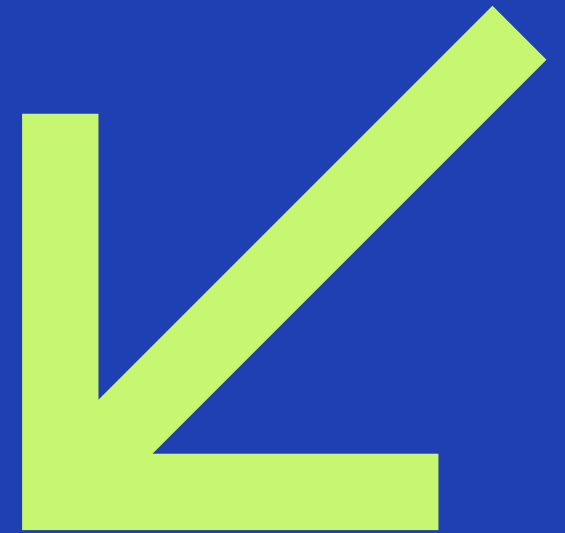
'Cuz he was holdin' it wrong





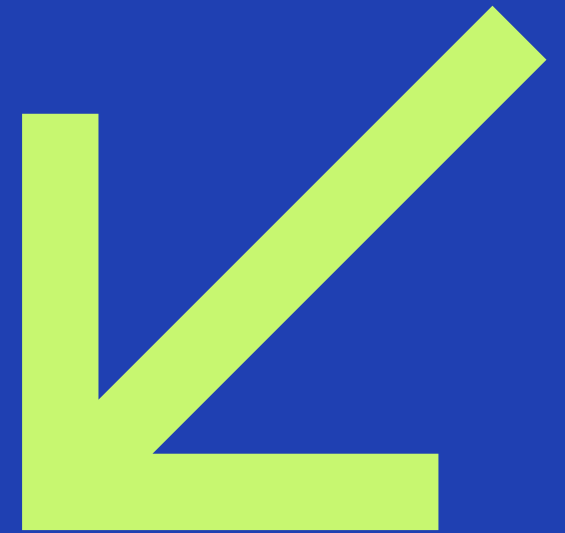
“

*In 2010 came the iPhone 4
25 per cent thinner, people queued out the door*



“

and a tempest in a teapot called Antennagate

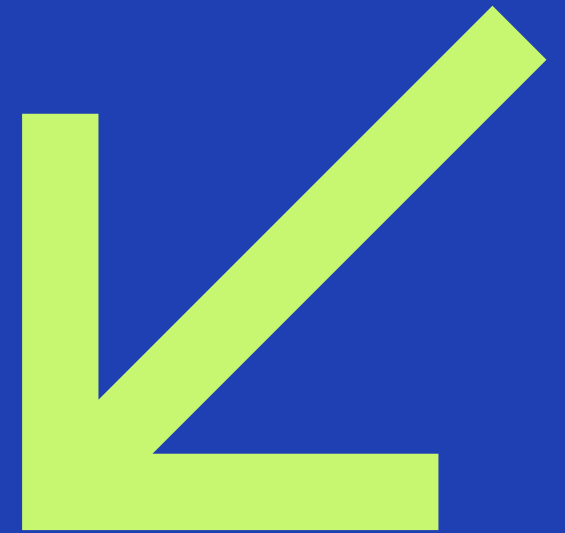


“

(someone asked)

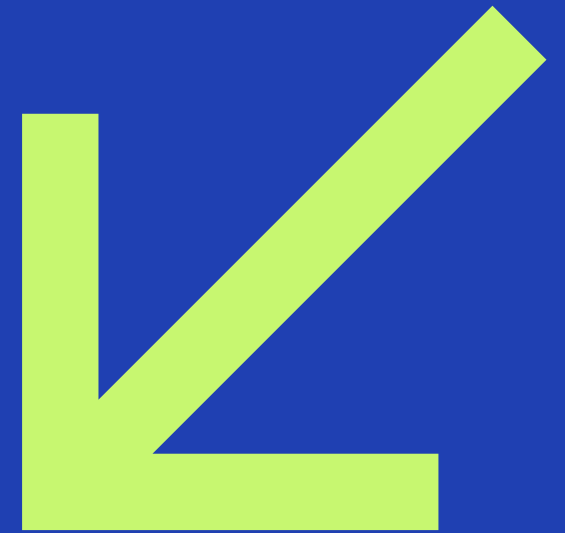
What's gonna be done 'bout my bars droppin'?

What? You're the one who is the signal blockin'



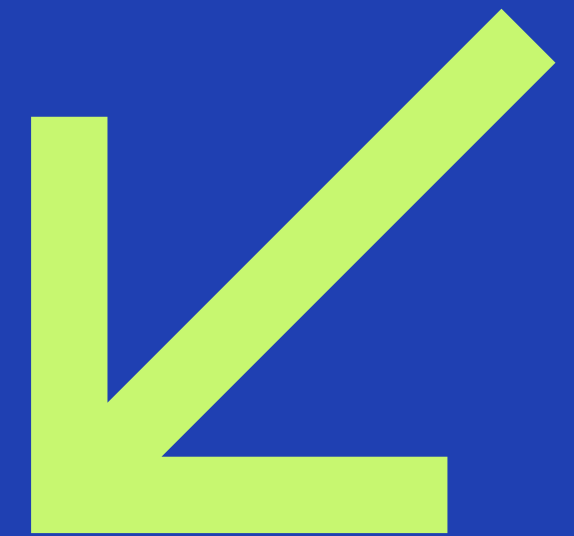
“

Steve Jobs had to set this poor sap straight



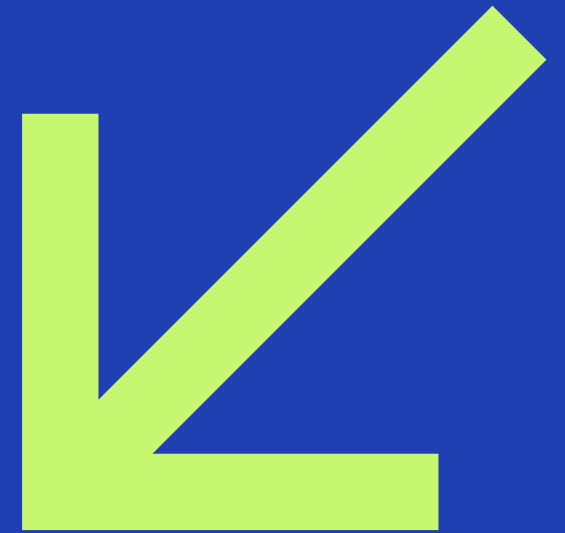
“

*He was holdin' it wrong
He was holdin' it wrong*



“

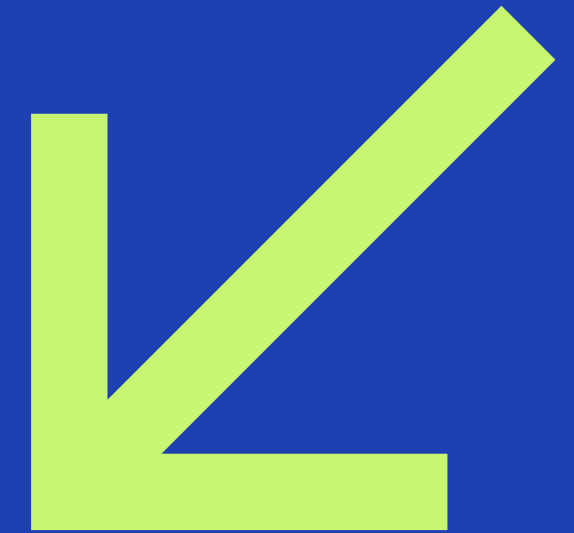
*Don't put yer hand there! Don't use a death grip!
Don't block that antenna! Oh, opinions let rip!
He was holdin' it wrong*





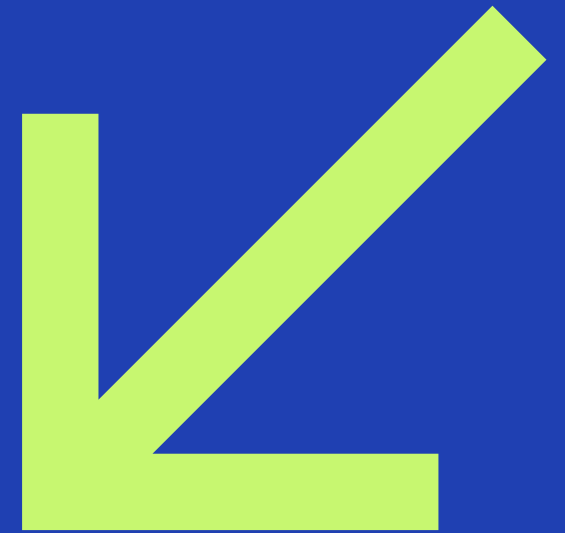
“

*Jimmy Bob's all confused 'bout PSSO
Still syncin' passwords - Ugh! I told him so!*



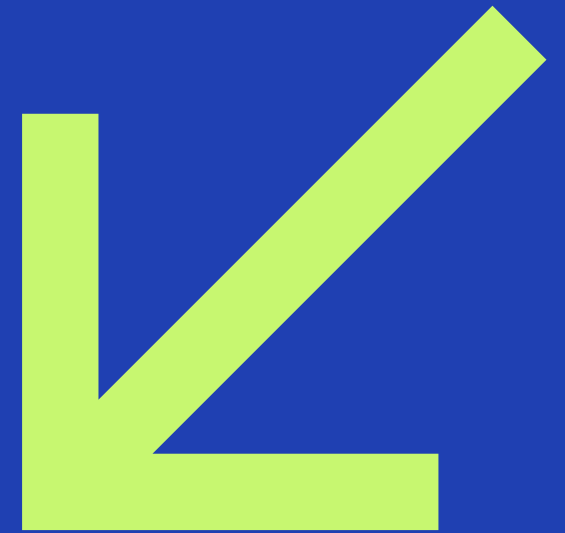
“

He's forever kickin' that tin can down the road



“

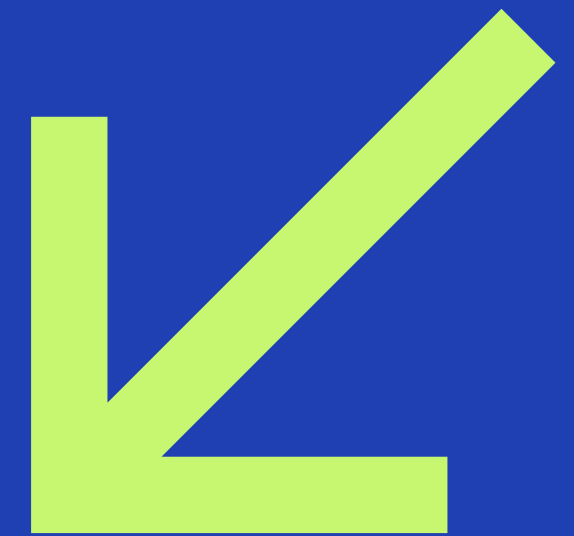
*Don't flush yer secur-ty straight down the crapper
yer tossin' the candy and eatin' the wrapper*



“

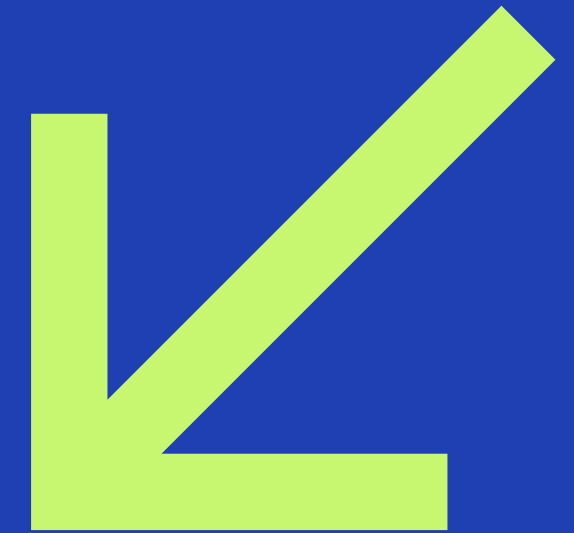
(besides)

Passkeys are far more convenient by the wagonload



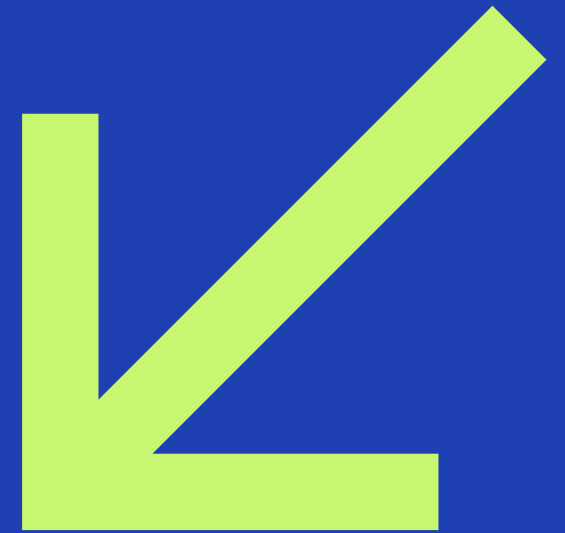
“

You're holdin' it wrong
You're holdin' it wrong



“

*Passwords ain't no way to secure your envir-ment
Secure Enclave must be the requir-ment
Or else you're holdin' it wrong*



“

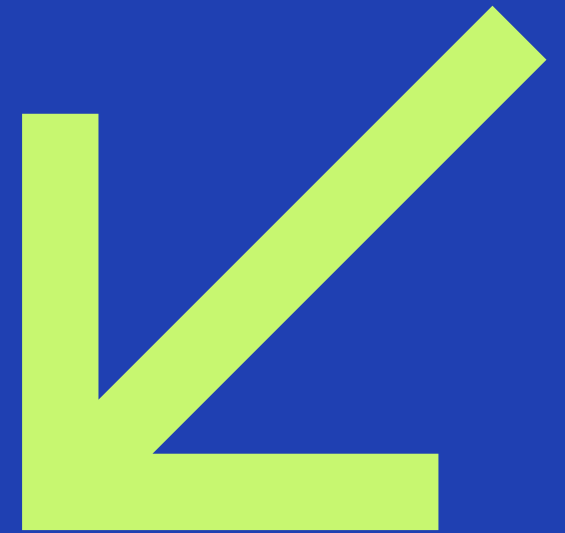
You're holdin' it wrong

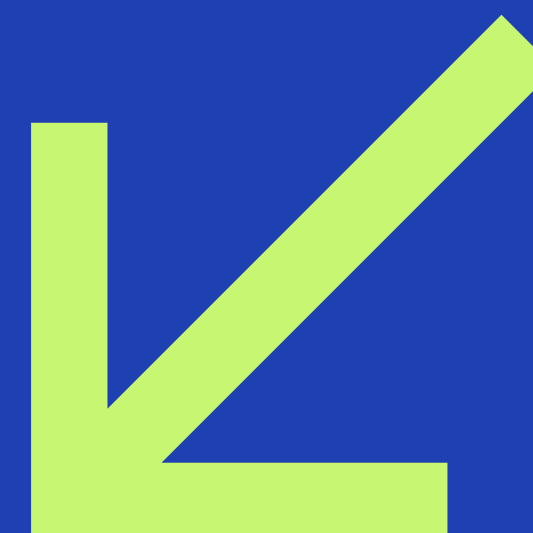
You're holdin' it wrong

Passwords ain't no way to secure your envir-ment

Secure Enclave must be the requir-ment

Or else you're holdin' it wrong





What Is the Deal With “Passwordless”?

This section will help you out



Macs Need Passwords

Yet...

**“Passwordless” is Microsoft's vision
for enterprise authentication**

**learn.microsoft.com shows 3,400+
results for “Passwordless”**

Apple Platform Security Guide?

One.

Conclusion: "Passwordless" is primarily a Microsoft term and has little or nothing to do with Macs or macOS

Does this get us anywhere?

What does Microsoft mean, when it says “Passwordless”?

**“Passwordless”
is short-hand for
(a shorter way to refer to)
“phishing-resistant
passwordless authentication”.**

**“Passwordless” is an overall
Security and Identity strategy
which specifies that
only phishing-resistant,
passwordless authentication
is used to access
Entra ID-protected resources.**

**Passwordless does *not* mean
that organizations merely offer
a safer, *alternative* means of
authentication
in addition to passwords.**

Passwordless does *not* mean
that organizations merely offer
a safer, *alternative* means of
authentication
in addition to passwords.

Passwordless means that
we do not use passwords at all
to access protected resources
in our environments!

Passwordless means that
we do not use passwords at all
to access **protected resources**
in our environments!

***We only use phishing-resistant
authentication***

**to access protected resources
in our environments!**

Passwords don't even exist!

Does passwordless apply to macOS?

No.

**Passwordless is primarily
a Microsoft term
and has little or nothing to do
with Macs or macOS**

**macOS is UNIX, and continues to
require a password across the OS**

How Macs Exist Within a Passwordless Microsoft Entra ID Environment

This section is the shortest, yet most important



So how and where do **Macs**,
which need passwords,
exist within a *passwordless*
Microsoft Entra ID *environment*?

**Macs must securely access
corporate resources
protected by
an identity provider (Entra ID)
using a phishing-resistant,
passwordless authentication factor**

**Macs must securely access
corporate resources
protected by
an identity provider (Entra ID)
using phishing-resistant,
passwordless authentication factors**

**We can literally turn our Macs
themselves into a
phishing-resistant, passwordless
authentication factor**

Understanding Passkeys

This section will *really* help you out



Passkeys are:

- ▶ **secure FIDO2 credentials**
- ▶ **specific to login.microsoft.com (Entra ID)**
- ▶ **up to 75% faster than passwords**

Two Types of Passkeys:

- ▶ **Device-bound OR**
- ▶ **Synchronized**

Synchronized Passkeys:

- ▶ **securely saved to a password or credential manager (Passwords.app)**
- ▶ **securely synchronized across your Apple devices (iCloud Keychain)**
- ▶ **examples: Google, Banks, Best Buy**

~~Synchronized Passkeys:~~

- ▶ ~~securely saved to a password or credential manager (Passwords.app)~~
- ▶ ~~securely synchronized across your Apple devices (iCloud Keychain)~~
- ▶ ~~examples: Google, Banks, Best Buy~~

Device-bound Passkeys:

- ▶ **never leave the device**
- ▶ **cannot be exported**
- ▶ **cannot be extracted**
- ▶ **example: YubiKey**
- ▶ **example: all your Macs**

**When macOS Platform SSO (PSSO)
is leveraged to create a Secure
Enclave-backed key (passkey),
it ensures that the Mac owner
is using an Entra ID joined Mac.**

**The Mac itself
becomes a hardware-based
authentication factor.**

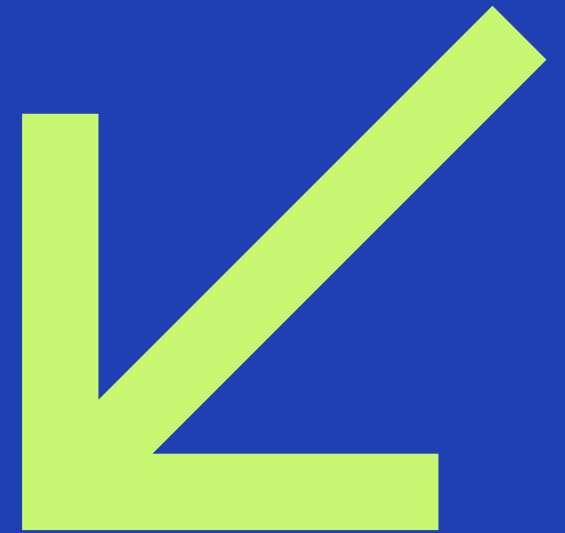
**There is only one way to create a
Secure Enclave-backed key
(passkey)**

***Hint:* You have to use Secure Enclave**

Feature	Secure Enclave		Password
Passwordless (phishing resistant) ←	✓		✗
TouchID supported for unlock	✓		✓
Can be used as passkey ←	✓		✗
MFA mandatory for setup	✓		✗
Multifactor authentication (MFA) is always recommended			
Local Mac password synced with Entra ID	✗		✓
Supported on macOS 13.x +	✓		✓
Supported on macOS 14.x +	✓		✓
Optionally, allow new users to log in with Entra ID credentials (macOS 14.x +)	✓		✓

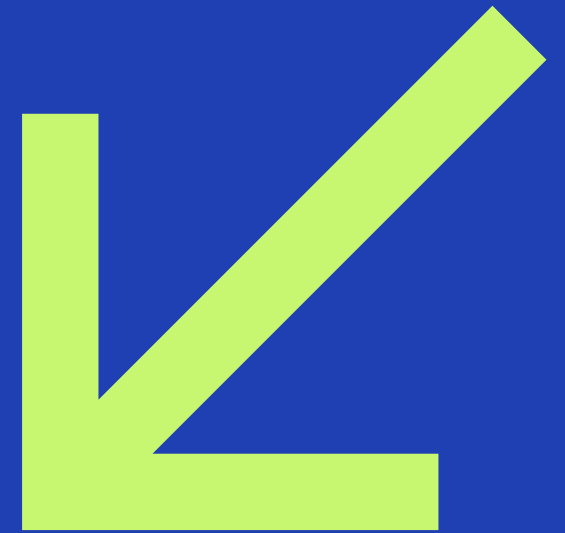
“

*Jimmy Bob's so confused 'bout P-SSO
Still synchin' passwords - Ugh! I told him so!
He's forever kickin' that tin can down the road*



“

*Don't flush yer secur-ty straight down the crapper
yer tossin' the candy and eatin' the wrapper
Passkeys are far more convenient by the wagonload*



Don't deploy Platform SSO with password authentication in 1:1 deployments

Changing from password authentication to Secure Enclave authentication - *which is inevitable* - requires a new Config Profile and your Mac users must re-register.

And what about syncing passwords to Entra ID?

No. Don't do it!

**You cannot sync a password
if it does not exist!**

**If you are still phasing Entra ID
passwords out (and they still exist),
don't use them on new Macs.**

Microsoft's recommendation?

Go back to using the MDM spec to manage password strength for the local Mac account

Use a passkey on an iPhone or smart phone to authenticate the Device Management enrollment

Microsoft's recommendation?

**Stop syncing passwords
over the network**

Is this all or nothing or is there a middle ground?

**What if my org./Security team/
Identity team is not ready to stop
synching passwords?**

You can deploy passkeys now and keep syncing passwords if you are already using Jamf Connect

You simply remove SSO and upgrade to Platform SSO to get passkeys

Determine when and how to stop syncing passwords in a future quarter

Passwords are hackable.

MFA is phishable.

**(without passkeys) your Org. is
vulnerable.**

Important Take-Aways

(this may get *spicy*)



**The entire point of PSSO is
passkeys.**

**The entire point of PSSO is
passkeys.**

Passkeys are the *what*.

Platform SSO is the *how*.

So who cares about PSSO?

The entire point of PSSO is passkeys.

Passkeys are the *what*.

Platform SSO is the *how*.

So who cares about PSSO?

**In fact, within your organization,
I suggest that you stop saying
"Platform SSO" or "PSSO" and
just say, "passkeys".**

How do Macs fit in a passwordless environment?

Passkeys.

How do we secure our orgs?

Passkeys.

What solution are we actually deploying?

Passkeys.

What ends up on your colleagues' Macs?

Passkeys.

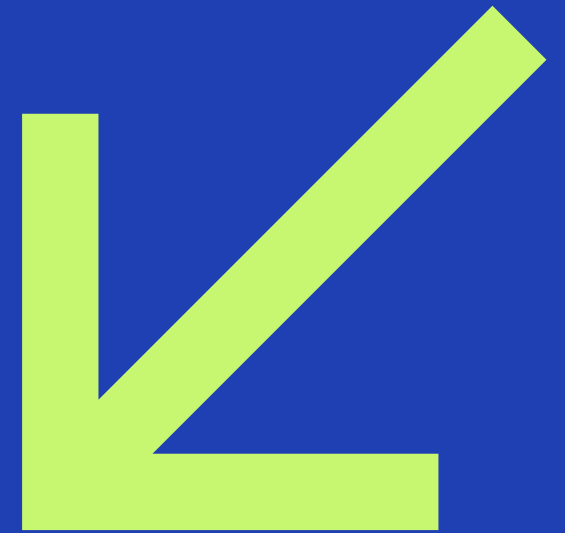
“

God is love.

Soylent Green is people.

Platform SSO is passkeys.

YOUR HUMBLE PRESENTER



How to Safely Test and Deploy Passkeys in Production



Consider This Progression

Passkeys are fairly easy to deploy if you are already deploying an SSO Extension Config. Profile

- ▶ Simply clone the SSO Extension Config. Profile, then change two settings to create a Platform SSO Config. Profile to deploy passkeys

Test and deploy the Platform SSO Config. Profile/ passkeys

- ▶ Passkeys can be deployed to your estate
- ▶ Passkeys work independently of Jamf Connect

Clone and update the Platform SSO Config. Profile for Simplified Setup

- ▶ Getting Simplified Setup to work is harder, but oh so cool!
- ▶ There are more moving parts (PreStage Enrollment, packages) and

Organizational Requirements

Your colleagues have/can install Microsoft Authenticator on their BYOD iPhones or smart phones

▶ Your colleagues have/can obtain a (device-bound) passkey

Your Org. has an authentication method policy in Microsoft Entra with Microsoft Authenticator and Passkey (FIDO2) enabled

Users must be able to join devices in Microsoft Entra

How to Safely Test and Deploy Passkeys in Production

Part I - To your Fleet



Requirements and Challenges

Requirements and Challenges - I

The SSO Extension Config. Profile and the PSSO Config. Profile cannot coexist

- ▶ We must remove the SSO Config. Profile, then deploy the Platform SSO Config. Profile

The Mac must *really should* restart after the SSO Extension Config. Profile is removed and before the Platform SSO Config. Profile is deployed

- ▶ We should not skip the restart. It removes old keychain items and cached files
- ▶ The PSSO Config. Profile and subsequent Entra ID registration also determines how compliance data flows through Jamf to Entra ID. Do you want to take chances with this?!

Requirements and Challenges - II

Because everyone hates restarting, the restart must be enforced

Therefore, IT must automate this workflow to create the best possible User Experience (UX)

The automation itself must span across/continue after the restart

The Mac needs to be running macOS 26 Tahoe to immediately trigger the Entra ID (re-)registration

Requirements and Challenges - II

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The Mac needs to be running macOS 26 Tahoe to immediately trigger the Entra ID (re-)registration

Requirements and Challenges



Requirements and Opportunities

The SSO Extension Config. Profile and the PSSO Config. Profile cannot coexist

- ▶ Brilliant Idea: What if removing the SSO Extension Config. Profile *triggers* the workflow?

Because everyone hates restarting, the restart must be enforced

- ▶ Idea: Let's use a Swift Dialog message to give our colleagues the ability to close their work
- ▶ Swift Dialog will remain on the screen until the user restarts

Requirements and Opportunities

macOS 26 Tahoe will immediately trigger the Entra ID (re-)registration

- ▶ Brilliant. Let's leverage that by making Tahoe a requirement!

Planning the Workflow - Components Needed

Planning the Workflow - Components Needed Before the Restart

**Removing the SSO Extension Config. Profile from target
Macs will trigger the workflow**

- ▶ We need a Smart Group to detect Macs without this profile. We'll call it "READY for PSSO Config"

**A Policy asking the user to restart will trigger at the next
checkin. The Policy consists of:**

- ▶ a script, which we will call "restart_for_passkey_set-up.sh" which
- ▶ displays a Swift Dialog message describing the passkey deployment steps and a button to restart.

Planning the Workflow - Components Needed After the Restart

Engineering Challenge #1 of 2:

- ▶ How do we automate the deployment of the “Platform SSO Extension” Config. profile as quickly as possible after a restart? Let's work through this logically, but backwards.
- ▶ The “Platform SSO Extension” Config. profile will be deployed to Macs in a Smart Group ([more on this later](#))
- ▶ We'll need to run a recon to get Macs to join this Smart Group
- ▶ We'll need a way to ensure the recon only runs for Macs in our workflow and because running too many recons is BAD...
- ▶ That recon must be triggered by a Policy for members of a (by definition) different Smart Group which no longer has an SSO profile and *needs* the PSSO profile

Planning the Workflow - Components Needed After the Restart

Engineering Challenge #2 of 2:

- ▶ If the “Platform SSO Extension” Config. Profile is scoped incorrectly, or deploys at the wrong time, TWO BAD THINGS could happen:
 - we could have an SSO Config. Profile and a PSSO Config. Profile on the Mac at the same time
 - the Entra ID registration will kick off before we are ready
- ▶ We'll need to create a Smart Group which has the following criteria:
 - the SSO Config. Profile has been removed
 - proof that the Macs are part of this workflow. But how?

Planning the Workflow - Components Needed Before the Restart

**Removing the SSO Extension Config. Profile from target
Macs will trigger the workflow**

- ▶ We need a Smart Group to detect Macs without this profile. We'll call it "READY for PSSO Config"

**A Policy asking the user to restart will trigger at the next
checkin. The Policy consists of:**

- ▶ a script, which we will call "restart_for_passkey_set-up.sh" which
- ▶ displays a Swift Dialog message describing the passkey deployment steps and a button to restart.
- ▶ creates ~/Library/PSSO_Staging/ folder (or any "meta-tag" of your choosing) to help trigger the PSSO Config. profile deployment

Planning the Workflow - Components Needed After the Restart

Engineering Challenge #2 of 2:

- ▶ If the “Platform SSO Extension” Config. Profile is scoped incorrectly, or deploys at the wrong time, TWO BAD THINGS could happen:
 - we could have an SSO Config. Profile and a PSSO Config. Profile on the Mac at the same time
 - the Entra ID registration will kick off before we are ready
- ▶ We'll need to create a Smart Group which has the following criteria:
 - the SSO Config. Profile has been removed
 - proof that the Macs are part of this workflow. But how?
- ▶ We'll also need an Extension Attribute to gather this proof/meta-tag, so that the recon which runs after the restart reads this proof

The Workflow

The Workflow - Before the Restart

- 1. Remove the SSO Extension Config. Profile from target Macs by adding specific groups to Scope > Exclusions (OR for Self Service version, add the Mac to a Static Group which is excluded)**
- 2. Macs without an SSO Extension Config. Profile join the Smart Group which we'll call "READY for PSSO Config" for the sake of this demo**

The Workflow - Before the Restart

3. Members of “READY for PSSO Config” are eligible for the Policy we'll call “Restart for Passkey Configuration” which triggers at the next checkin. The Policy runs:
 - a. the “restart_for_passkey_set-up.sh” script which
 - i. creates the ~/Library/PSSO_Staging/ folder (or any “meta-tag” of your choosing)
 - ii. displays a Swift Dialog message describing the passkey deployment steps and the need to restart.

[the user restarts the Mac by clicking "Restart Now" -
the only button in the SD message]

The Workflow - After the Restart

4. After the restart, a special recon will run, as part of a Policy we'll call "Inventory Collection-STARTUP-for PSSO". During this recon:
 - a. an Extension Attribute looks for the presence of the ~/Library/PSSO_Staging/ folder
 - b. Macs which have this folder join a Smart Group we'll call "READY for PSSO Config. Profile" Smart Group to which the "AUTH-MS Platform SSO Extension" Config. Profile will be deployed

The Workflow - After the Restart

5. The “AUTH-MS Platform SSO Extension” Config. Profile deploys soon after reaching the Desktop
6. The deployment of the “AUTH-MS Platform SSO Extension” Config. Profile triggers an Entra ID registration, which, as part of the PSSO settings, includes a passkey.
7. The user follows the prompts which are built into macOS

What our Colleagues See:

Restart Required for Passkey Setup



Passkeys for macOS

Before we can generate a passkey, your Mac requires a restart.

Please save your work then click 'Restart Now'.

Thanks for keeping Sophos safe!

Restart Now

Beware:

The "Register" button will not appear unless the user moves the cursor over the macOS notification.

Make sure users and the Help Desk know this in advance.



Registration Required

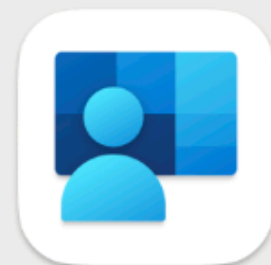
Please register with your identity provider.



Registration Required

Please register with your identity p

Register



Platform Single Sign-on Registration



Device Registration

Register your Mac computer with Sophos.

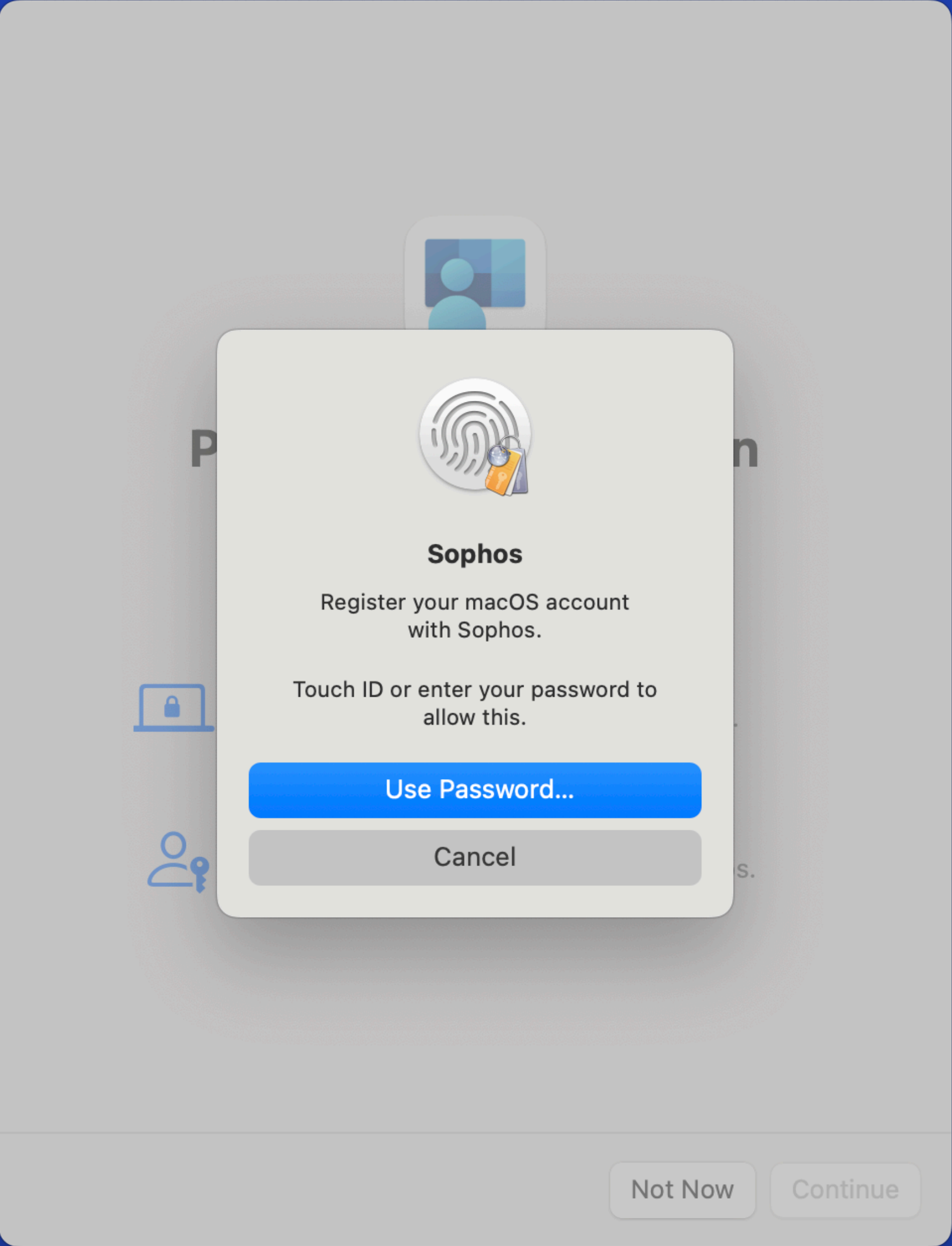


Account Registration

Register your macOS account with Sophos.

Not Now

Continue



Register your device with Microsoft Entra



Sign in

Email, phone, or Skype

No account? [Create one!](#)

[Can't access your account?](#)

Back

Next



Sign-in options

[Terms of use](#)

[Privacy & cookies](#)

...

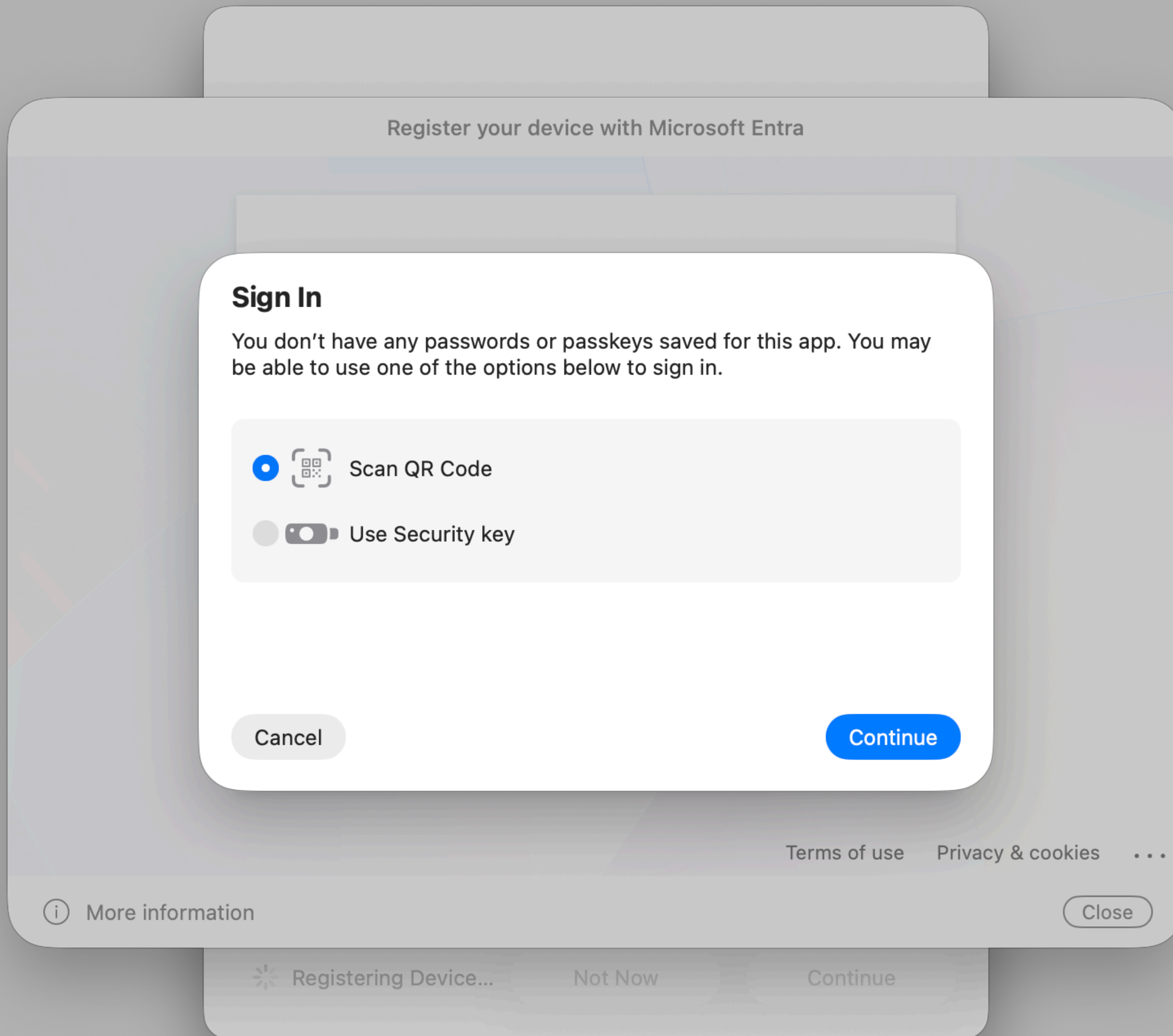
[More information](#)

Close

Registering Device...

Not Now

Continue





**Use your other device to choose a passkey for
"Company Portal".**

Cancel

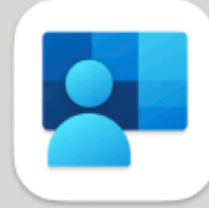
 More information

Close

 Registering Device...

Not Now

Continue



Enable your Entra ID passkey

To use your Entra ID passkey, you must enable Company Portal as a Passkey Provider.

To complete this action, open the System Settings app and navigate to:
General > Autofill & Passwords > Autofill from > Enable Company Portal.

AutoFill Passwords and Passkeys

Automatically suggest passwords, passkeys, and verification codes when signing in to apps and websites.



AutoFill from



Passwords

Passkeys, passwords, and codes



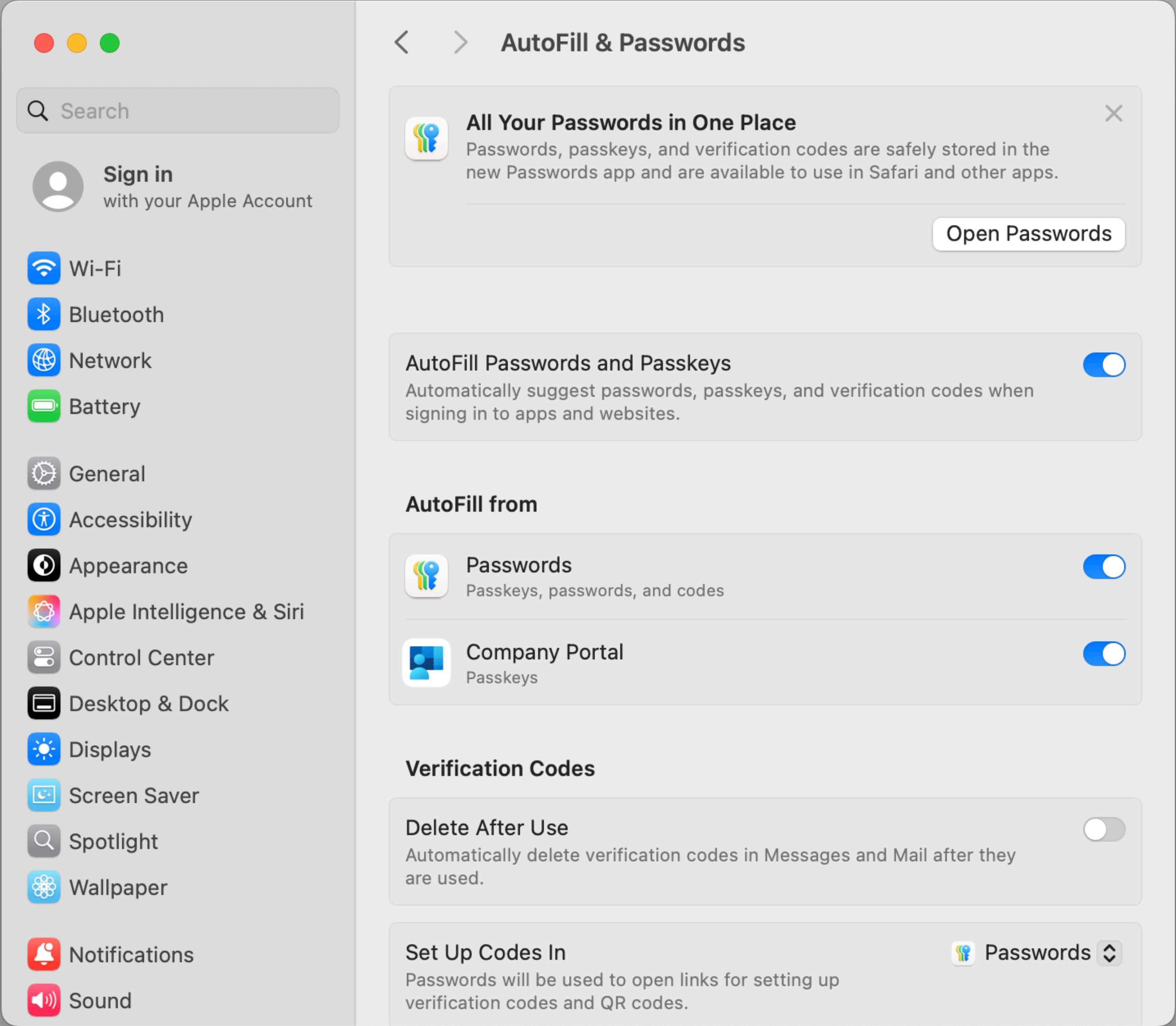
Company Portal

Passkeys



Open System Settings

Dismiss

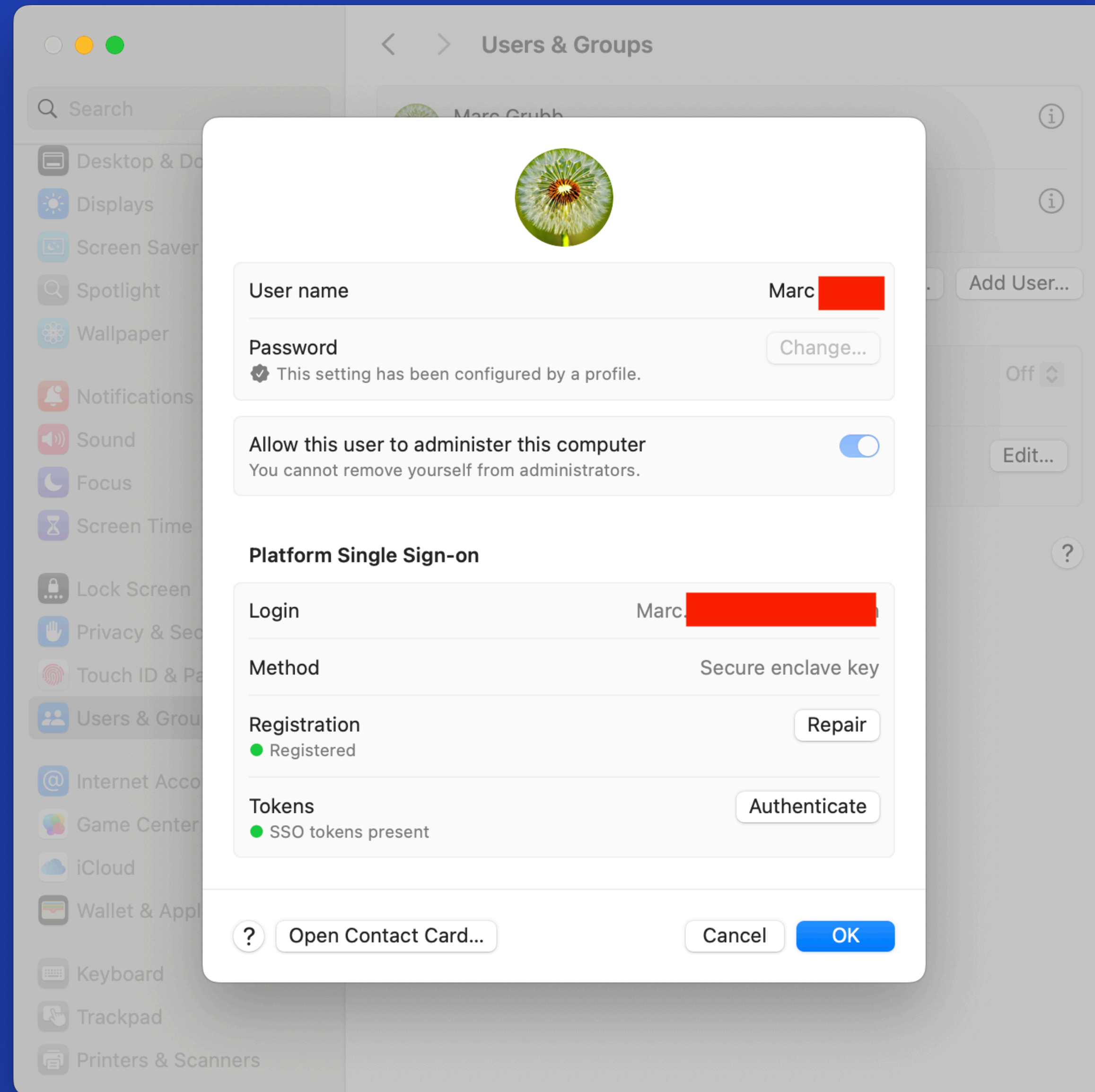




Platform Single Sign-on Registration

Registration Complete

Close



How to Safely Test and Deploy Passkeys in Production

Part II - New Mac Builds & Re-builds



Two Methods:

1. Account creation *and* a passkey but *no* password sync using Platform SSO and Simplified Setup (*before* login) (most secure)
2. Account creation *and* password sync using Jamf Connect, *then* passkey deployment (*after* login)

Two Methods:

(what your colleagues encounter during ADE)

Simplified Setup

- ▶ PSSO screen
- ▶ MS Auth screens (passkey)
- ▶ Account & passkey created *invisibly* (PSSO)
- ▶ Password creation (colleague)
- ▶ Setup Asst. *screens*
- ▶ (Desktop)

Jamf Connect

- ▶ JC Account creation
- ▶ Corporate password entry
- ▶ Setup Asst. *screens*
- ▶ (Desktop)
- ▶ Entra ID Registration (notification)
- ▶ Single Sign-On for Mac *screens*
- ▶ MS Auth screens (passkey)
- ▶ Company Portal Autofill message
- ▶ Autofill & Passwords (System Settings)
- ▶ Registration Complete screen

Two Methods:

(what your colleagues encounter during ADE)

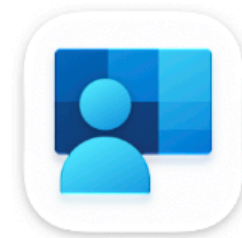
Simplified Setup

- ▶ PSSO screen
- ▶ MS Auth screens (passkey)
- ▶ Account & passkey created *invisibly* (PSSO)
- ▶ Password creation (colleague)
- ▶ Setup Asst. *screens*
- ▶ (Desktop)

Jamf Connect

- ▶ JC Account creation
- ▶ Corporate password entry
- ▶ Setup Asst. *screens*
- ▶ (Desktop)
- ▶ Entra ID Registration (notification)
- ▶ Single Sign-On for Mac screens
- ▶ MS Auth screens (passkey)
- ▶ Company Portal Autofill message
- ▶ Autofill & Passwords (System Settings)
- ▶ Registration Complete screen

What our Colleagues See:



Single Sign-On for Mac

With single sign-on, you only have to sign in once to securely access your Mac and apps.



Automatically Registered

Your Mac and macOS account are registered to use single sign-on with identity provider.



Access to Apps and Data

You can access your organization's apps and personal data when your Mac is unlocked.

Continue

What our Colleagues See:

Create a Mac Account

Your name and account name have been determined by your organization. The password you create here will be used to log in to this Mac.

Marc

Marc@.com

This will be the name of your home folder.

Hint (Optional)

Continue

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Deployment Tips

Beware:

The "Register" button will not appear unless the user moves the cursor over the macOS notification.

Make sure your colleagues and the Help Desk know this in advance.



Registration Required

Please register with your identity provider.



Registration Required

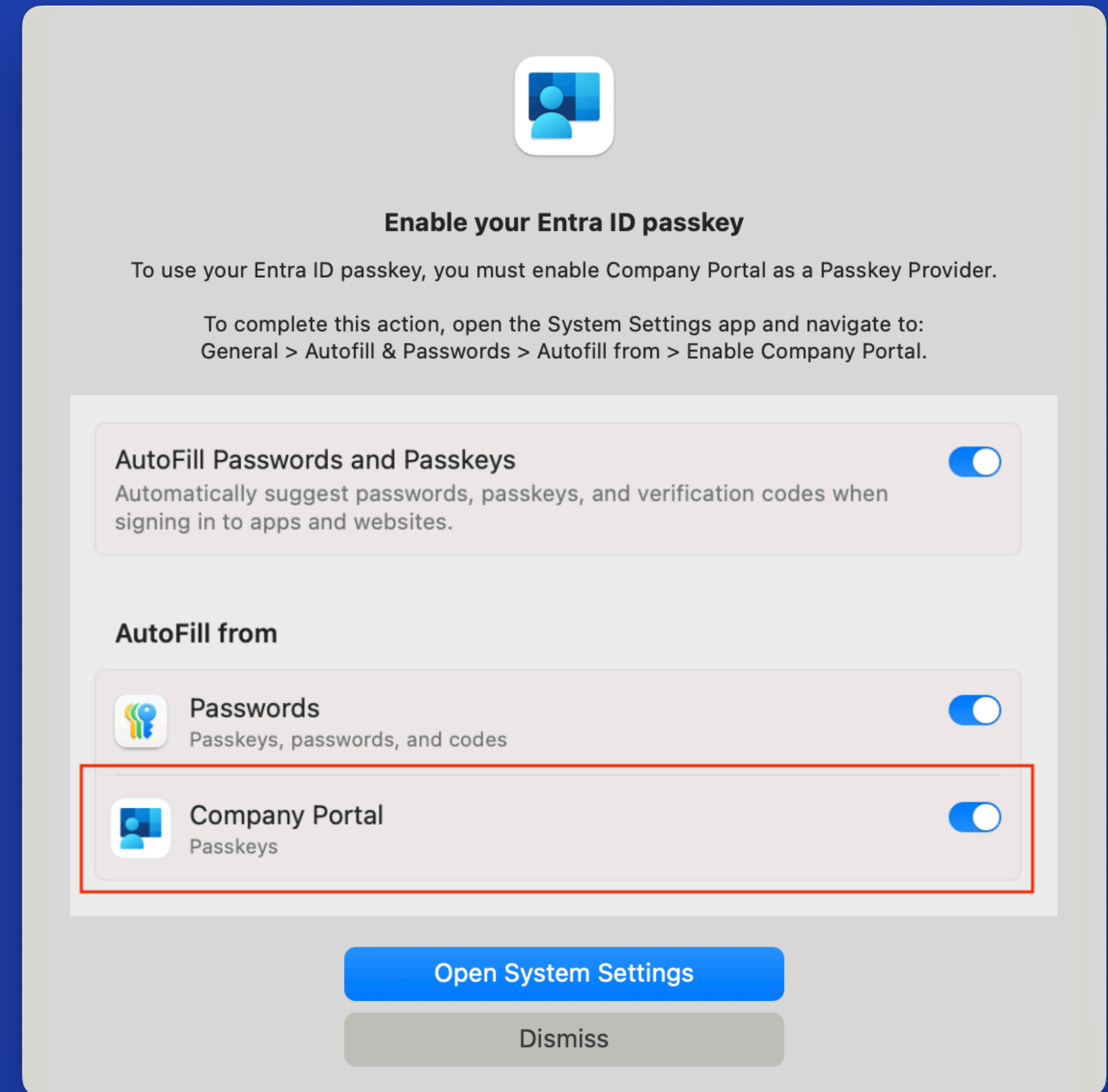
Please register with your identity p

Register

Beware:

The "Autofill & Passwords" message does not appear in the Simplified Setup workflow.

Make sure your colleagues and the Help Desk know this in advance.



Browser Support:

- ▶ **Safari: Apple-native support**
- ▶ **Edge: Microsoft-native support**
- ▶ **Chrome: Enterprise version supported**
- ▶ **Firefox: Deploy a Config Profile**
- ▶ **Chromium-based: it depends...
(may need an extension)**

Private/Incognito Windows:

If your Identity Team *also* checks for compliance *in addition to* requiring authentication for specific websites or resources, the compliance check will want to set a cookie, and *it will fail*.

(Even worse) Entra ID It assumes the Mac is not enrolled in Jamf, and will prompt a Mac registration.

Comms:

- ▶ **Don't use the same comms for Windows and Macs**
- ▶ **Make sure you explain the difference between "device-bound" passkeys and synchronized passkeys.**
- ▶ **Don't compare with YubiKeys which must be "registered"**

Working with your Identity Team:

**Your Identity Team will need to
require passwordless auth. in its
Conditional Access Rules (CARs)**

(Ask them, not me)

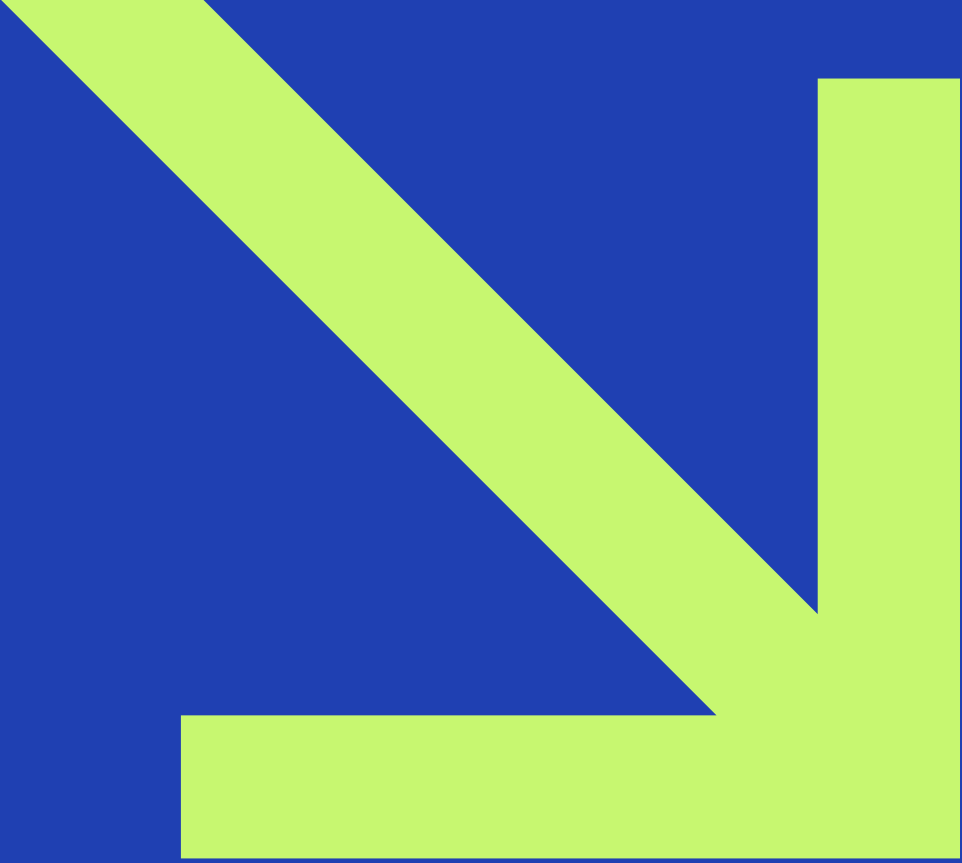
Resources:

<https://github.com/ka-pow-er/>

#platform-sso



QUESTIONS?
YOUR TURN!



THANK
YOU